MODULE 09

# Azure Deployment and Operations

Day 2 · Optional MVP Module

**ADOBE STOCK PLACEHOLDER**

*Suggested search: “Azure cloud dashboard on a monitor (wide hero banner)”*

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# LEARNING OBJECTIVES

- Provision Azure resources with repeatable templates.
- Configure identity, monitoring, and alerting.
- Validate runtime telemetry and operational readiness.

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*Suggested search: “cloud computing data center (supporting image)”*

# WHY THIS MODULE MATTERS

This module extends the core track into cloud operations.

It is optional for MVP delivery when quota or tenant constraints are present.

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*Suggested search: “IT operations monitoring wall of screens (concept image)”*

# CONCEPT MAP

*How eShopOnWeb's own infra/main.bicep turns into a running, observable environment.*

1. **BICEP (IaC)**
   Declarative main.bicep defines the environment

2. **AZURE RESOURCES**
   App Service, Key Vault, Azure SQL provisioned

3. **APPLICATION INSIGHTS**
   Requests, dependencies, exceptions telemetry

4. **ALERTS**
   Actionable signals on top of the telemetry

# FROM THE DOCS: BICEP, PRECISELY

- Bicep is a declarative domain-specific language for deploying Azure resources.
- Bicep is a transparent abstraction over ARM JSON — the Bicep CLI transpiles it to ARM at deployment.
- Benefit Microsoft states: Bicep files are more concise and easier to read than the equivalent ARM JSON.
- Benefit Microsoft states: Bicep gets day-one support for new/preview Azure resource types.
- Benefit Microsoft states: no state files to manage — Azure itself stores deployment state.

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*Suggested search: “cloud security and identity concept (detail image)”*

*Source: What is Bicep? — Azure Resource Manager — https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/overview*

# COURSEWARE FOCUS

- Azure hosting options
- Identity
- Observability
- Bicep
- Costs

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# REAL IAC — ESHOPONWEB INFRA/MAIN.BICEP

### REAL CODE — infra/main.bicep (excerpt)

```
targetScope = 'subscription'

param environmentName string
param location string

@secure()
param sqlAdminPassword string

resource rg 'Microsoft.Resources/resourceGroups@2021-04-01' = {
  name: !empty(resourceGroupName) ? resourceGroupName
    : '${abbrs.resourcesResourceGroups}${environmentName}'
  location: location
  tags: tags
}

module web './core/host/appservice.bicep' = {
  name: 'web'
  scope: rg
  params: { runtimeName: 'dotnetcore', runtimeVersion: '8.0' }
}
```

*Provisions the App Service, Key Vault, and two Azure SQL databases this module's lab deploys.*

# TOOLS & RESOURCES

## IaC Templates

**Azure Quickstart Templates / Bicep files (Azure/azure-quickstart-templates)**

*Infrastructure as Code deployment for App Service, AKS, and Azure SQL.*

## Monitoring Docs

**Microsoft Learn: Monitor application performance with Application Insights**

*Telemetry, logging, and SLA dashboard setup.*

## Migration Guides

**Azure SQL Database Migration Guides (Data Migration Assistant / Azure Migrate)**

*Database migration hands-on reference.*

# SUPPORTING ASSETS

- Azure templates and scripts from local references
- App artifacts from prior modules

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*Suggested search: “cloud computing data center (reference screenshot)”*

# LAB 09: AZURE DEPLOYMENT AND OPERATIONS

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*Suggested search: “IT operations monitoring wall of screens (lab / hands-on photo)”*

# LAB STEPS

Goal: Deploy to Azure and establish baseline observability.

Tools: Azure CLI, Bicep · Requires an Azure subscription with quota.

- Step 1 — Deploy core resources from templates.
- Step 2 — Deploy the app artifact.
- Step 3 — Configure identity and secret access.
- Step 4 — Enable logs, metrics, and alerts.
- Step 5 — Validate app and telemetry behavior.
- Validation: deployment succeeds without manual rework, and monitoring/alerts show active signals.

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*Suggested search: "cloud security and identity concept (hands-on lab photo)"*

# KNOWLEDGE CHECK: Azure Deployment and Operations

**Which Azure capability is used to write infrastructure as repeatable, version-controlled templates?**

A. Azure Bicep

B. Azure Front Door

C. Azure DevTest Labs

D. Azure Cost Management

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**ADOBE STOCK PLACEHOLDER**

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# **KNOWLEDGE CHECK:** Azure Deployment and Operations

**Per Microsoft's own docs, what does the Bicep CLI do with a .bicep file at deployment time?**

A. Executes it directly as PowerShell

B. Transpiles it into an ARM JSON template

C. Converts it to Terraform

D. Nothing — no conversion happens

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# SOURCES & FURTHER READING

## EXTERNAL DOCUMENTATION

- **What is Bicep? — Azure Resource Manager**
  https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/overview

## REFERENCED IN THIS REPOSITORY

- `course/repos/eShopOnWeb/infra/main.bicep`
- `course/repos/eShopOnWeb/azure.yaml`

# RECAP & SUCCESS CRITERIA

- Workload is deployed and observable with actionable alerts.
- Tier: Optional MVP Module

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